

Week 10 — Time-Frequency Analysis

Homework, Weekly Quiz, and Solutions

Lectures 19–20: DFT / Fast Convolution • STFT / Spectrograms • DSP-to-ML bridge

Homework Problems

1. DFT grid and interpretation

A 64-sample record is acquired at $F_s = 8$ kHz. (a) What is the DFT bin spacing in Hz? (b) At what frequency is bin $k = 10$? (c) If the record is zero-padded to a 256-point FFT, what is the displayed bin spacing? Has the fundamental frequency resolution improved?

2. Linear versus circular convolution

Let $x[n] = \{1, 2, 1, -1\}$ and $h[n] = \{1, -1, 2\}$. (a) Compute the linear convolution. (b) State the minimum DFT length required for DFT multiplication to reproduce it exactly. (c) Explain what happens if a 4-point DFT is used instead.

3. FFT convolution design

A streaming FIR has 129 taps. You use overlap-add with $B = 384$ new input samples per block. (a) What is the minimum FFT length? (b) Which power-of-two FFT length would be a natural choice? (c) How many output samples from each block convolution can be nonzero before overlap-add?

4. Overlap-save bookkeeping

An overlap-save implementation uses $N_{\text{FFT}} = 1024$ and an FIR of length 65. (a) How many old input samples must overlap between adjacent blocks? (b) How many new output samples are retained per FFT block? (c) Which samples are discarded and why?

5. STFT parameter reasoning

A vibration signal sampled at 12 kHz contains 20-ms transient events and two steady tones separated by 40 Hz. Compare STFT windows of 128, 512, and 2048 samples. Discuss which gives better time localization, which gives better frequency separation, and why there is no universally best choice.

6. Spectrogram interpretation

Sketch or describe the spectrogram appearance of (a) a constant sinusoid, (b) a short impulse-like transient, (c) a linear chirp, and (d) a periodic voiced signal with harmonics.

7. PyTorch tensor design

A batch contains 32 real audio clips. An STFT produces 257 one-sided frequency bins and 101 time frames per clip. (a) Give a suitable Conv2d input shape. (b) Explain why arbitrary 90° rotations of these “images” are usually a poor augmentation. (c) Give two DSP-informed augmentations that could be more sensible.

8. Integrated system design

Design a real-time classifier front end for a 16-kHz sensor stream. It needs an FIR noise-reduction filter followed by a spectrogram and CNN, with an end-to-end preprocessing latency below 40 ms. Propose reasonable FIR block-processing and STFT parameters and justify the trade-offs. There is no single correct numeric design; the reasoning must be consistent.

Homework Solutions

1. DFT grid and interpretation

(a) $\Delta f = F_s/N = 8000/64 = 125$ Hz. (b) $f_{10} = 10(125) = 1250$ Hz. (c) A 256-point zero-padded FFT has grid spacing $8000/256 = 31.25$ Hz. This makes the plotted DTFT sampling denser, but the observation length is still 64 samples, so the underlying resolving power of the finite record/window has not improved.

2. Linear versus circular convolution

Linear convolution: $\{1, 1, 1, 2, 0, -2\}$. The required length is $4 + 3 - 1 = 6$, so $\text{NDFT} \geq 6$. With $N = 4$, the samples at linear-convolution indices 4 and 5 wrap modulo 4 and add into output indices 0 and 1, producing time aliasing.

3. FFT convolution design

Minimum $\text{NFFT} = B + L_h - 1 = 384 + 129 - 1 = 512$. A 512-point FFT is therefore already a natural power-of-two choice. Each block linear convolution has $B + L_h - 1 = 512$ potentially nonzero samples. Adjacent block outputs overlap by $L_h - 1 = 128$ samples and are added.

4. Overlap-save bookkeeping

Overlap is $L_h - 1 = 64$ input samples. Valid new outputs per block are $\text{NFFT} - L_h + 1 = 960$. Discard the first 64 IFFT output samples because circular wrap-around contaminates them; concatenate the remaining 960 samples.

5. STFT parameter reasoning

At 12 kHz, $128/12000 \approx 10.7$ ms, $512/12000 \approx 42.7$ ms, and $2048/12000 \approx 170.7$ ms. The 128-sample window best localizes 20-ms events but has poor frequency separation. The 2048-sample window best separates nearby tones but strongly smears short transients. The 512-sample window is a plausible compromise. A strong answer explicitly recognizes that required time localization and tone separation conflict.

6. Spectrogram interpretation

(a) Constant sinusoid: a horizontal ridge at the sinusoid frequency. (b) Short transient: a narrow vertical stripe spanning many frequencies. (c) Linear chirp: a sloped ridge. (d) Voiced harmonic signal: a stack of approximately horizontal harmonic ridges at integer multiples of the fundamental, possibly changing with time.

7. PyTorch tensor design

(a) $[32, 1, 257, 101]$ is a standard Conv2d shape: batch, channel, frequency, time. (b) Rotation swaps/reverses time and frequency relationships and usually has no physical meaning. (c) Examples include additive noise, gain changes, small time shifts, limited time masking, limited frequency masking, plausible time stretching, or pitch changes when appropriate to the application.

8. Integrated system design

One defensible design: use overlap-save with a 129-tap FIR and $\text{NFFT} = 512$, retaining 384 new samples = 24 ms per block. Use a 256-sample (16-ms) Hann STFT window with hop 64 samples (4 ms). This gives relatively low temporal latency and dense frame updates while still providing moderate frequency resolution. A different design can earn full credit if it explicitly budgets filter block delay, FIR group delay, STFT windowing delay, computation, and model latency under the 40-ms requirement.

Homework Grading Guidance

Problem	Points	What earns full credit
1	10	Correct bin calculations; clearly distinguishes zero-padding from

		true resolution.
2	12	Correct linear convolution, padding requirement, and wrap-around explanation.
3	10	Correct NFFT and overlap-add lengths.
4	10	Correct overlap, retained sample count, and discard rationale.
5	14	Quantitative window durations plus time/frequency trade-off reasoning.
6	10	Correct time–frequency pattern descriptions.
7	14	Correct tensor shape and domain-aware augmentation reasoning.
8	20	Coherent latency-aware design with defensible DSP/ML parameter choices.

Weekly Quiz

1. An N-point DFT multiplication of two sequences implements what kind of convolution by default?
2. A length-50 sequence is convolved with a length-17 FIR by FFT. What minimum transform length avoids time aliasing?
3. In one sentence, distinguish overlap-add from overlap-save.
4. An STFT uses a 256-sample window at $F_s = 8$ kHz. What is the window duration in milliseconds?
5. Which generally improves frequency localization: a longer or shorter STFT window? What does it cost?
6. What quantity is visualized by a power spectrogram?
7. What spectrogram pattern would you expect from an impulse-like transient?
8. Why can spectrograms be natural inputs to Conv2d layers?

Weekly Quiz Solutions

1. N-point circular convolution.
2. $50 + 17 - 1 = 66$ points minimum.
3. Overlap-add uses non-overlapping input blocks and adds overlapping output tails; overlap-save overlaps input blocks and discards the contaminated first $L_h - 1$ output samples.
4. $256/8000 = 0.032$ s = 32 ms.
5. A longer window generally improves frequency localization/resolution but worsens time localization and can smear fast changes.
6. The squared magnitude of the STFT, $|X(r, \omega)|^2$, usually displayed on a linear or dB scale.

7. A narrow vertical broadband stripe because the event is localized in time and contains a wide range of frequencies.

8. The spectrogram is a 2-D structured array over frequency and time, so local convolutional kernels can learn nearby time–frequency patterns such as ridges, harmonics, and transients.

Differentiation for a Mixed First-Year / Senior Cohort

First-year minimum target: correctly identify DFT versus FFT, linear versus circular convolution, the role of zero-padding, the basic OLA/OLS idea, and how to read common spectrogram patterns.

Senior extension target: derive block-validity lengths, reason about latency/throughput, distinguish true time–frequency resolution from display density, and justify ML preprocessing parameters as part of a reproducible signal-processing pipeline.

Good answers should state the principle first, then support it with a calculation, diagram, or physical interpretation. Avoid rewarding memorized parameter values without a reason for choosing them.